seeed studio

INDUSTRIAL PRODUCT DATASHEET

reComputer Robotics J3011 with GMSL Extension - Intelligent Edge AI Computer with NVIDIA Jetson Orin Nano Super 8GB

reComputer Robotics J3011 with GMSL Extension: NVIDIA Jetson-powered edge AI computer with up to 157 TOPS, GMSL2 connectivity, and industrial-grade design for advanced robotics.

SKU	UPDATED	PRICE
100001302	2026-10-27 04:14:49	$819.00(Excl. VAT)



Overview



The reComputer Robotics J3011 with GMSL Extension - Intelligent Edge AI Computer with NVIDIA Jetson Orin Nano Super 8GB delivers up to 157 TOPS AI performance by NVIDIA Jetson Orin NX 16GB module in Super/MAXN mode, with rich robotics interfaces including dual RJ45, 6x USB 3.2, 2x CAN, GMSL2, I2C, and UART. It offers industrial-grade power and thermal design (–20 °C to 60 °C, 19–54 V DC), pre-installed JetPack 6.2, and ready-to-use GMSL extension for applications such as humanoid robots, robotic arms, AMRs, and vision-based robotics.

Key Features

01 **Boosted AI Performance in a compact size**

Delivers up to 157 TOPS AI performance by NVIDIA® Jetson™ Orin™ NX 16GB module in Super/MAXN mode.

02 **Rich Interfaces for Robotics**

Including dual RJ45, M.2 slots for 5G/Wi-Fi/BT modules, 6x USB 3.2, 2x CAN, GMSL2, I2C, and UART, functioning as a powerful robotic brain.

03 **Industrial-Grade Power and Thermal Design**

Operates reliably from –20 °C to 60 °C in 25 W mode and up to 55 °C in MAXN mode, fueled through a wide 19–54 V DC input, ensuring stable operation in mobile and industrial robot environments

04 **Software-ready Ecosystem**

Pre-installed with JetPack 6.2 and Linux BSP for seamless deployment. Combined with Isaac Sim, it can be used to create autonomous mobile robots (AMRs), arms and manipulators, humanoids, and more.

05 **Ready-to-useGMSLExtension**

Provides ready-to-useGMSLextensionthatsupportsmultipleGMSLcamerasinJetpack

06 **Jetson Modules Performance with Super Mode**

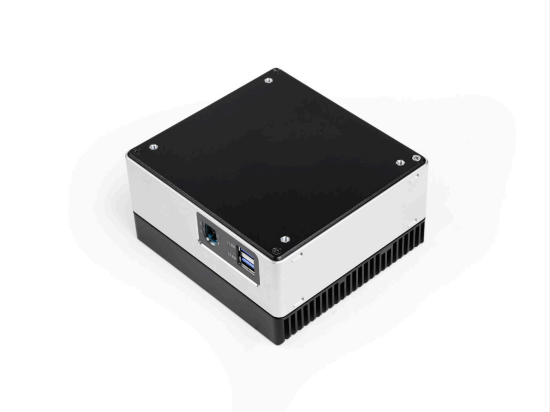
07 **Enhanced power management**

08 **Improved higher frame rate processing**

09 **Reduced inference time**

Product Gallery





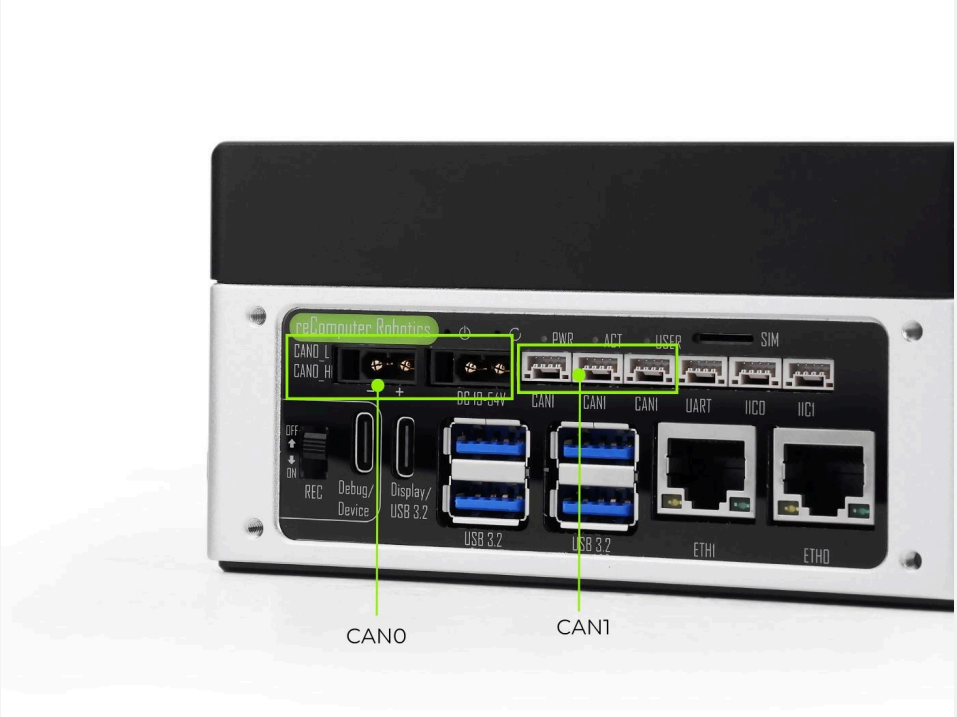


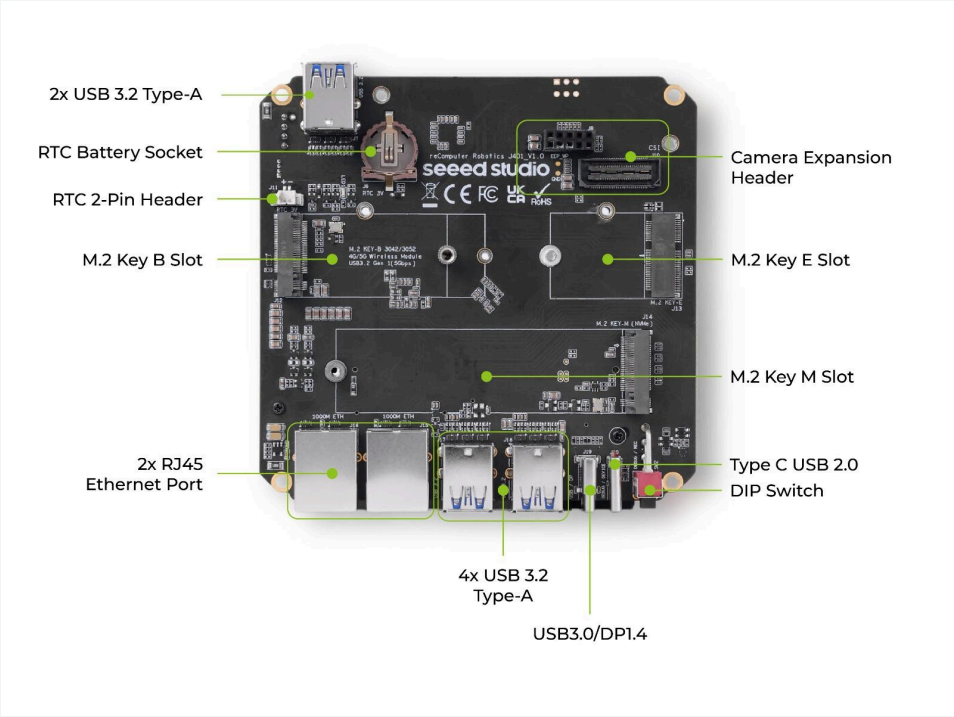






Hardware Overview





Applications

- **Humanoid Robot & Manipulation:** Serving as a powerful brain for humanoid robots, the reComputer Robotics series delivers rich perception with precise motion control. Emulating human-like sensing, dexterity, mobility and whole-body coordination—seamlessly collaborating with people to get tasks done.
- **CAN bus links servos or motors:** The CAN bus links servos or motors to drive Robotic Arms and Autonomous Robots with millisecond-level joint coordination.
- **Logistics Robots map via UART LiDAR:** Logistics Robots map in seconds via UART LiDAR, dodge on the fly and shuttle racks nonstop.
- **I²C sensor data net:** I²C weaves temperature, humidity, IMU and fuel-gauge data into a net, giving robots early-maintenance alerts.
- **USB barcode scanning and ticket printer:** USB cameras scan barcodes while on the move, and a USB ticket printer labels parcels in real time—zero manual labor from inbound to outbound.
- **Dual-RJ45 link-aggregation video:** Dual-RJ45 link-aggregation delivers video to the Edge AI reCopmputer in <1 ms for real-time QC and remote monitoring.
- **High-Performance Vision for Robotics:** Combining a GMSL 2 camera with a reComputer Robotics enables advanced vision-based robotics. Sensing 3MP GMSL 2 camera supports real-time object detection, panoramic environment awareness, and multi-camera 3D reconstruction . The high-bandwidth, low-latency GMSL 2 link supports long cable runs and multi-camera setups, while the Jetson Orin NX delivers real-time AI processing for perception, mapping, and control.
- **3D Vision and GMSL 2 Connectivity for Robotics:** The reComputer Robotics GMSL Extension supports the Orbbec Gemini 335LG Stereo Vision 3D Camera, providing accurate real-time 3D perception for advanced robotics. With the MAX96712 deserializer and mini-FAKRA connector, it ensures secure, low-latency, and reliable data transmission, ideal for AMRs, robotic arms, and collaborative robots. Precise depth sensing enables smarter navigation, robust object recognition, and reliable obstacle avoidance across mobile robots, delivery systems, and shelf-stocking applications.

Resource Links

- https://www.seeedstudio.com/reComputer-Robotics-J3011-with-GMSL-extension-board-p-6538.html
- User Manual&Datasheet
- Power Board Schematic
- GMSL Board Schematic
- 3D File
- Mechanical Document-reComputer Robotics PCBA
- Nvidia Jetson Comparison
- Seeed Nvidia Jetson Success Cases
- Seeed Jetson One Pager
- Flash BSP for Jetson

Seeed Technology Co.,Ltd.
9F, Building G3, International E City,
Zhongshanyuan Road, Nanshan, Shenzhen, China

Tel: +86 0755-80695676
Web: www.seeed.cc
Shop: www.seeedstudio.com